

Hypothesis Testing: One Sample

MDragon Data Tools - Study Guide

What is Hypothesis Testing?

A statistical method to make decisions about population parameters based on sample data.

The Five-Step Process

Step 1: State Hypotheses

- H_0 (Null Hypothesis): Status quo, no effect, equality (=)
- H_a (Alternative Hypothesis): What we're trying to prove ($\neq$, $<$, $>$)

Step 2: Choose Significance Level

- α (alpha): Probability of Type I error
- Common: $\alpha = 0.05, 0.01, 0.10$

Step 3: Calculate Test Statistic

Z-test (σ known, $n \geq 30$):

$$z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$$

T-test (σ unknown, small n):

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$
$$df = n - 1$$

Step 4: Find P-Value or Critical Value

- Two-tailed: $H_a: \mu \neq \mu_0$
- Right-tailed: $H_a: \mu > \mu_0$
- Left-tailed: $H_a: \mu < \mu_0$

Step 5: Make Decision

- If p-value $\leq \alpha$: Reject H_0
- If p-value $> \alpha$: Fail to reject H_0

Example 1: Right-Tailed Z-Test

 **Problem:** A company claims their product weighs 500g on average ($\sigma = 20$ g). A sample of 40 products has $\bar{x} = 508$ g. Test at $\alpha = 0.05$.

 **Solution:**

1. $H_0: \mu = 500$; $H_a: \mu > 500$ (right-tailed)
2. $\alpha = 0.05$
3. $z = (508 - 500) / (20/\sqrt{40}) = 8 / 3.16 = 2.53$
4. p-value = $P(Z > 2.53) = 0.0057$
5. $0.0057 < 0.05$, **Reject H_0**

Conclusion: Significant evidence that mean weight exceeds 500g.

Example 2: Two-Tailed T-Test

 **Problem:** Test if mean = 75 when sample of $n=16$ has $\bar{x}=78$, $s=8$, $\alpha=0.05$.

 **Solution:**

1. $H_0: \mu = 75$; $H_a: \mu \neq 75$
2. $\alpha = 0.05$
3. $t = (78-75)/(8/\sqrt{16}) = 3/2 = 1.5$, $df=15$

4. $p\text{-value} = 2 \times P(T > 1.5) \approx 0.155$

5. $0.155 > 0.05$, **Fail to reject H_0**

Conclusion: No significant evidence that mean differs from 75.

Type I and Type II Errors

Decision	H_0 True	H_0 False
Reject H_0	Type I Error (α)	Correct!
Fail to Reject	Correct!	Type II Error (β)

Practice with our Hypothesis Test Calculator!

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mondragon-developer.github.io/statools